

Problem

Find the Fourier transform of the following signals.

(a) $f_1(t) = e^{-3t} \sin(10t)u(t)$

(b) $f_2(t) = e^{-4t} \cos(10t)u(t)$

Step-by-step solution

Step 1 of 9

(a)

The Fourier transform of a signal is,

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Substitute $e^{-3t} \sin(10t) \cdot u(t)$ for $f(t)$. It exists for $t > 0$ and dies down at $t = \infty$.

The signal is a converging signal.

Thus, the Fourier transform exists for the signal

Divide the signal $e^{-3t} \sin(10t) \cdot u(t)$ into two parts; $e^{-3t}u(t)$ and $\sin(10t)$.

The actual signal is the multiplication of these two signals in time domain.

Step 2 of 9

The Fourier transform of $e^{-3t}u(t)$ is given as follows:

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} e^{-3t} u(t) e^{-j\omega t} dt \\ &= \int_0^{\infty} e^{-(3+j\omega)t} dt \\ &= \left[\frac{e^{-(3+j\omega)t}}{-(3+j\omega)} \right]_0^{\infty} \quad \left(\text{Since } \int e^{-ax} = \frac{e^{-ax}}{-a} \right) \\ &= \frac{1}{3+j\omega} \quad \left(\text{Since } e^{-\infty} = 0 \right) \end{aligned}$$

Step 3 of 9

The Fourier transform of impulse function is,

$$\delta(t) \leftrightarrow 1$$

The Fourier transform of a shifted impulse function is,

$$\delta(t - t_0) \leftrightarrow e^{j\omega t_0}$$

Duality property is given as follows:

$$X(t) \leftrightarrow 2\pi x(-\omega) \dots\dots (1)$$

Determine the Fourier transform of 1.

Use duality property.

Therefore, the Fourier transform of 1 is,

$$\begin{aligned} F(\omega) &= 2\pi\delta(-\omega) \\ &= 2\pi\delta(\omega) \quad (\text{Since Impulse function is even, } \delta(-\omega) = \delta(\omega)) \end{aligned}$$

Determine the Fourier transform of $e^{j\omega_0 t}$.

$$\delta(t - t_0) \leftrightarrow e^{j\omega t_0}$$

$$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega_0 - \omega) \quad (\text{By duality property})$$

$$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0) \quad (\text{Since Impulse function is even, } \delta(\omega_0 - \omega) = \delta(\omega - \omega_0))$$

Thus, the Fourier transform of $e^{j\omega_0 t}$ is,

$$e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega - \omega_0) \dots\dots (2)$$

Step 4 of 9

Determine the Fourier transform of $\sin(10t)$.

$$\begin{aligned} X_1(\omega) &= \int_{-\infty}^{\infty} \sin(10t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} \left(\frac{e^{j10t} - e^{-j10t}}{2j} \right) e^{-j\omega t} dt \end{aligned}$$

Use equation (2) for further simplification.

$$\begin{aligned} X_1(\omega) &= \int_{-\infty}^{\infty} \left(\frac{e^{j10t} e^{-j\omega t} - e^{-j10t} e^{-j\omega t}}{2j} \right) dt \\ &= \frac{2\pi\delta(\omega - 10) - 2\pi\delta(\omega + 10)}{2j} \\ &= \frac{\pi\delta(\omega - 10) - \pi\delta(\omega + 10)}{j} \end{aligned}$$

Thus the Fourier transform of $\sin(10t)$ is, $X_1(\omega) = \left(\frac{\pi\delta(\omega - 10) - \pi\delta(\omega + 10)}{j} \right)$

Step 5 of 9

Multiplication in time domain is convolution in frequency domain.

$$f(t) \cdot p(t) \leftrightarrow \frac{1}{2\pi} (F(\omega) * P(\omega))$$

Since the two signal parts are multiplied in time domain, convolute them in frequency domain.

Determine the Fourier transform of the signal.

$$\begin{aligned} &= \frac{1}{2\pi} \left(\frac{\pi \delta(\omega - 10) - \pi \delta(\omega + 10)}{j} * \frac{1}{3 + j\omega} \right) \\ &= \frac{1}{2\pi} \left(\frac{\pi}{j} \left(\frac{1}{3 + j(\omega - 10)} - \frac{1}{3 + j(\omega + 10)} \right) \right) \quad (\text{Sifting property: } \delta(x - x_0) * P(x) = P(x_0)) \\ &= \frac{1}{2\pi} \left(\frac{\pi}{j} \left(\frac{20j}{(3 + j\omega)^2 + 100} \right) \right) \\ &= \left(\frac{10}{(3 + j\omega)^2 + 100} \right) \end{aligned}$$

Thus, the Fourier transform of the signal is,

$$F_1(\omega) = \left(\frac{10}{(3 + j\omega)^2 + 100} \right)$$

Step 6 of 9

(b)

The Fourier transform of a signal is,

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

Substitute $e^{-4t} \cos(10t) \cdot u(t)$ for $f(t)$. It exists for $t > 0$ and dies down at $t = \infty$.

The signal is a converging signal.

Thus, the Fourier transform exists for the signal.

Divide the signal $e^{-4t} \cos(10t) \cdot u(t)$ into two parts; $e^{-4t} u(t)$ and $\cos(10t)$.

The actual signal is the multiplication of these two signals in time domain.

Step 7 of 9

The Fourier transform of $e^{-4t} u(t)$ is given as follows:

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} e^{-4t} u(t) e^{-j\omega t} dt \\ &= \int_0^{\infty} e^{-(4+j\omega)t} dt \\ &= \left[\frac{e^{-(4+j\omega)t}}{-(4+j\omega)} \right]_0^{\infty} \quad \left(\text{Since } \int e^{-ax} = \frac{e^{-ax}}{-a} \right) \\ &= \frac{1}{4+j\omega} \quad (\text{Since } e^{-\infty} = 0) \end{aligned}$$

Step 8 of 9

Determine the Fourier transform of $\cos(10t)$.

$$\begin{aligned} X_1(\omega) &= \int_{-\infty}^{\infty} \cos(10t) e^{-j\omega t} dt \\ &= \int_{-\infty}^{\infty} \left(\frac{e^{j10t} + e^{-j10t}}{2} \right) e^{-j\omega t} dt \end{aligned}$$

Use equation (2) for further simplification.

$$\begin{aligned} X_1(\omega) &= \int_{-\infty}^{\infty} \left(\frac{e^{j10t} e^{-j\omega t} + e^{-j10t} e^{-j\omega t}}{2} \right) dt \\ &= \frac{2\pi\delta(\omega - 10) + 2\pi\delta(\omega + 10)}{2} \\ &= \pi\delta(\omega - 10) + \pi\delta(\omega + 10) \end{aligned}$$

Thus, the Fourier transform of $\cos(10t)$ is, $X_1(\omega) = \pi\delta(\omega - 10) + \pi\delta(\omega + 10)$

Step 9 of 9

Multiplication in time domain is convolution in frequency domain.

$$f(t) \cdot p(t) \leftrightarrow \frac{1}{2\pi} (F(\omega) * P(\omega))$$

Since the two signal parts are multiplied in time domain, convolute them in frequency domain.

Determine the Fourier transform of the signal.

$$\begin{aligned} &= \frac{1}{2\pi} \left((\pi\delta(\omega - 10) + \pi\delta(\omega + 10)) * \frac{1}{4 + j\omega} \right) \\ &= \frac{1}{2\pi} \left(\pi \left(\frac{1}{4 + j(\omega - 10)} + \frac{1}{4 + j(\omega + 10)} \right) \right) \quad (\text{Sifting property: } \delta(x - x_0) * P(x) = P(x_0)) \\ &= \frac{1}{2\pi} \left(\pi \left(\frac{2(4 + j\omega)}{(4 + j\omega)^2 + 100} \right) \right) \\ &= \left(\frac{4 + j\omega}{(4 + j\omega)^2 + 100} \right) \end{aligned}$$

Thus, the Fourier transform of the signal is,

$$F_2(\omega) = \left(\frac{4 + j\omega}{(4 + j\omega)^2 + 100} \right)$$